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CO-TENANT ALERT APPARATUS, SYSTEM, AND METHOD

Abstract

A method is disclosed for alerting a co-tenant of a fault in a partitioned system. An apparatus and system also perform the functions of the method. The method includes monitoring, for faults, two or more partitions in a system sharing a management controller where each of the partitions is associated with a tenant. The method includes detecting a fault on a first partition and identifying an alert class of multiple alert classes for the fault. The method includes identifying the alert class of the fault as a high-level alert class indicating that the fault affects one or more other partitions different than the first partition. The method includes notifying the tenant of the first partition and each tenant of the one or more other partitions of the fault in response to the alert class being the high-level alert class.

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Background/Summary

FIELD

[0001] The subject matter disclosed herein relates to managing computing systems and more particularly relates to managing partitioned computing systems.

BACKGROUND

[0002] In contemporary computing systems, partitioning is a commonly employed technique to enhance reliability, security, and overall system efficiency. Partitioned systems consist of two or more partitions, each operating independently and serving specific functions within the broader system. These partitions may run on the same processor or on distinct processors, depending on the system architecture.

[0003] While partitioning provides numerous advantages, it also introduces challenges related to fault detection and response. In a partitioned system, faults or errors occurring in one partition have the potential to impact other partitions. Therefore, there is a need for an effective mechanism to detect fault events in one partition and promptly alert the processor associated with another partition.

BRIEF SUMMARY

[0004] A method is disclosed for alerting a co-tenant of a fault in a partitioned system. An apparatus and system also perform the functions of the method. The method includes monitoring, for faults, two or more partitions in a system sharing a management controller where each of the partitions is associated with a tenant. The method includes detecting a fault on a first partition and identifying an alert class of multiple alert classes for the fault. The method includes identifying the alert class of the fault as a high-level alert class indicating that the fault affects one or more other partitions different than the first partition. The method includes notifying the tenant of the first partition and each tenant of the one or more other partitions of the fault in response to the alert class being the high-level alert class.

[0005] According to another aspect of the disclosure, an apparatus includes one or more processors and a non-transitory computer readable storage media storing code. The code is executable by the one or more processors to perform operations including monitoring, for faults, each of two or more partitions in a system sharing a management controller. Each of the one or more partitions is associated with a tenant. The operations further include detecting a fault on a first partition of the two or more partitions and identifying an alert class of multiple alert classes for the fault. The operations include identifying that the alert class of the fault is a high-level alert class where the high-level alert class indicates that the fault affects one or more of the two or more partitions different than the first partition. The operations further include notifying the tenant of the first partition and each tenant of the one or more other partitions of the fault in response to the alert class being a high-level alert class.

[0006] According to another aspect of the present disclosure, a system includes a partitionable motherboard with two or more partitions and a management controller configured to communicate with the two or more partitions. The management controller is connected to tenants of the two or more partitions over a management network. The system includes one or more processors on the management controller and non-transitory computer readable storage media storing code. The code is executable by the one or more processors to perform operations including monitoring, for faults, each of two or more partitions in a system sharing a management controller where each of the two or more partitions is associated with a tenant. The operations further include detecting a fault on a first partition of the two or more partitions and identifying an alert class of multiple alert classes for the fault. The operations include identifying that the alert class of the fault is a high-level alert class, where the high-level alert class indicates that the fault affects one or more other partitions of the two or more partitions different than the first partition. The operations further include notifying

the tenant of the first partition and each tenant of the one or more other partitions of the fault in response to the alert class being a high-level alert class.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] A more particular description of the embodiments briefly described above will be rendered by reference to specific embodiments that are illustrated in the appended drawings. Understanding that these drawings depict only some embodiments and are not therefore to be considered to be limiting of scope, the embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings, in which:

[0008] FIG. 1 is a schematic block diagram illustrating a system for monitoring two or more partitions for faults according to various embodiments;

[0009] FIG. 2 is a schematic block diagram illustrating a system for detecting and responding to faults in a partitioned system having two tenants, according to various embodiments;

[0010] FIG. 3 is a schematic block diagram illustrating another system for monitoring partitions for faults where the partitions are in a partitioned compute node, according to various embodiments;

[0011] FIG. 4 is a schematic block diagram illustrating an apparatus for monitoring and responding to faults in a partitioned system, according to various embodiments;

[0012] FIG. 5 is a schematic block diagram illustrating another apparatus for monitoring and responding to faults in a partitioned system, according to various embodiments;

[0013] FIG. 6 is a schematic block diagram illustrating a plurality of alert classes, according to various embodiments;

[0014] FIG. 7 is a schematic flow chart diagram illustrating one embodiment of a method for monitoring and responding to faults in a partitioned system, according to various embodiments; and

[0015] FIG. 8 is a schematic flow chart diagram illustrating another embodiment of a method for monitoring and responding to faults in a partitioned system, according to various embodiments.

DETAILED DESCRIPTION

[0016] As will be appreciated by one skilled in the art, aspects of the embodiments may be embodied as a system, method or program product. Accordingly, embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Furthermore, embodiments may take the form of a program product embodied in one or more computer readable storage devices storing machine readable code, computer readable code, and/or program code, referred to hereafter as code. The storage devices, in some embodiments, are tangible, non-transitory, and/or non-transmission.

[0017] Many of the functional units described in this specification have been labeled as modules, in order to more particularly emphasize their implementation independence. For example, a module may be implemented as a hardware circuit comprising custom very large scale integrated (“VLSI”) circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. A module may also be implemented in programmable hardware devices such as a field programmable gate array (“FPGA”), programmable array logic, programmable logic devices or the like.

[0018] Modules may also be implemented in code and/or software for execution by various types of processors. An identified module of code may, for instance, comprise one or more physical or logical blocks of executable code which may, for instance, be organized as an object, procedure, or function. Nevertheless, the executables of an identified module need not be physically located

together, but may comprise disparate instructions stored in different locations which, when joined logically together, comprise the module and achieve the stated purpose for the module.

[0019] Indeed, a module of code may be a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, and across several memory devices. Similarly, operational data may be identified and illustrated herein within modules, and may be embodied in any suitable form and organized within any suitable type of data structure. The operational data may be collected as a single data set, or may be distributed over different locations including over different computer readable storage devices. Where a module or portions of a module are implemented in software, the software portions are stored on one or more computer readable storage devices.

[0020] Any combination of one or more computer readable medium may be utilized. The computer readable medium may be a computer readable storage medium. The computer readable storage medium may be a storage device storing the code. The storage device may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, holographic, micromechanical, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. A computer readable storage medium, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

[0021] More specific examples (a non-exhaustive list) of the storage device would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (“RAM”), a read-only memory (“ROM”), an erasable programmable read-only memory (“EPROM” or Flash memory), a portable compact disc read-only memory (“CD-ROM”), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

[0022] Code for carrying out operations for embodiments may be written in any combination of one or more programming languages including an object oriented programming language such as Python, Ruby, R, Java, Java Script, Smalltalk, C++, C sharp, Lisp, Clojure, PHP, or the like, and conventional procedural programming languages, such as the “C” programming language, or the like, and/or machine languages such as assembly languages. The code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (“LAN”) or a wide area network (“WAN”), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0023] Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment, but mean “one or more but not all embodiments” unless expressly specified otherwise. The terms “including,” “comprising,” “having,” and variations thereof mean “including but not limited to,” unless expressly specified otherwise. An enumerated listing of items does not imply that any or all of the items are mutually exclusive, unless expressly specified otherwise. The terms “a,” “an,” and “the” also refer to “one or more” unless expressly specified otherwise.

[0024] Furthermore, the described features, structures, or characteristics of the embodiments may be combined in any suitable manner. In the following description, numerous specific details are

provided, such as examples of programming, software modules, user selections, network transactions, database queries, database structures, hardware modules, hardware circuits, hardware chips, etc., to provide a thorough understanding of embodiments. One skilled in the relevant art will recognize, however, that embodiments may be practiced without one or more of the specific details, or with other methods, components, materials, and so forth. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of an embodiment.

[0025] Aspects of the embodiments are described below with reference to schematic flowchart diagrams and/or schematic block diagrams of methods, apparatuses, systems, and program products according to embodiments. It will be understood that each block of the schematic flowchart diagrams and/or schematic block diagrams, and combinations of blocks in the schematic flowchart diagrams and/or schematic block diagrams, can be implemented by code. This code may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the schematic flowchart diagrams and/or schematic block diagrams block or blocks.

[0026] The code may also be stored in a storage device that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the storage device produce an article of manufacture including instructions which implement the function/act specified in the schematic flowchart diagrams and/or schematic block diagrams block or blocks.

[0027] The code may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer implemented process such that the code which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0028] The schematic flowchart diagrams and/or schematic block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of apparatuses, systems, methods and program products according to various embodiments. In this regard, each block in the schematic flowchart diagrams and/or schematic block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions of the code for implementing the specified logical function(s).

[0029] It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. Other steps and methods may be conceived that are equivalent in function, logic, or effect to one or more blocks, or portions thereof, of the illustrated Figures.

[0030] Although various arrow types and line types may be employed in the flowchart and/or block diagrams, they are understood not to limit the scope of the corresponding embodiments. Indeed, some arrows or other connectors may be used to indicate only the logical flow of the depicted embodiment. For instance, an arrow may indicate a waiting or monitoring period of unspecified duration between enumerated steps of the depicted embodiment. It will also be noted that each block of the block diagrams and/or flowchart diagrams, and combinations of blocks in the block diagrams and/or flowchart diagrams, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and code.

[0031] The description of elements in each figure may refer to elements of proceeding figures. Like

numbers refer to like elements in all figures, including alternate embodiments of like elements.

[0032] As used herein, a list with a conjunction of “and/or” includes any single item in the list or a combination of items in the list. For example, a list of A, B and/or C includes only A, only B, only C, a combination of A and B, a combination of B and C, a combination of A and C or a combination of A, B and C. As used herein, a list using the terminology “one or more of” includes any single item in the list or a combination of items in the list. For example, one or more of A, B and C includes only A, only B, only C, a combination of A and B, a combination of B and C, a combination of A and C or a combination of A, B and C. As used herein, a list using the terminology “one of” includes one and only one of any single item in the list. For example, “one of A, B and C” includes only A, only B or only C and excludes combinations of A, B and C.

[0033] As previously discussed, existing solutions often fail to recognize and appropriately respond to faults that may impact more than one partition in a partitioned system. The present disclosure addresses this and other issues by introducing a novel method for detecting faults in a partitioned system and alerting a processor associated with a specific partition when the fault event is detected in another partition. This innovation ensures a proactive and efficient response to faults in a partitioned system, thereby minimizing an impact on system performance.

[0034] The method includes monitoring, for faults, two or more partitions in a system sharing a management controller, where each of the partitions is associated with a tenant. The method includes detecting a fault on a first partition and identifying an alert class of multiple alert classes for the fault. The method includes identifying the alert class of the fault as a high-level alert class indicating that the fault affects one or more other partitions different than the first partition. The method includes notifying the tenant of the first partition and each tenant of the one or more other partitions of the fault in response to the alert class being the high-level alert class.

[0035] In some embodiments, the high-level alert class includes multiple sub-classes. Each of the sub-classes is related to a response time within which a remediation action must be performed. The method may further include identifying an alert sub-class corresponding to the fault in response to identifying that the alert class of the fault is a high-level alert class.

[0036] In some embodiments, the method includes determining, based on the identified sub-class, a remediation action to be performed and/or timing of the remediation action. In some embodiments, the remediation action includes, for the first partition and the one or more other partitions, moving data, shedding load, refusing critical loads, and redirecting new data to a non-shared drive.

[0037] In some embodiments, the method further includes, based on the identified sub-class, immediately performing, at the first partition and/or the one or more other partitions, the remediation action associated with the sub-class. The method may include, based on the identified sub-class, scheduling, at the first partition and/or the one or more other partitions, the remediation action associated with the sub-class for a future time. In some embodiments, in response to the identified alert class including the sub-class for scheduling the remediation action at a future time and in response to detecting a second fault with an identified alert class comprising a high-level alert class, the method further includes changing the scheduled remediation to an earlier time.

[0038] In certain embodiments, the method further includes building a map of hardware components related to more than one of the two or more partitions. In some embodiments, identifying the alert class for the fault and identifying that the alert class is a high-level alert class further includes accessing information from the map to determine if a component related to the fault is related to more than one of the two or more partitions.

[0039] In some embodiments, notifying the tenants of the one or more other partitions includes communicating, over a management network, the fault to the tenants of the one or more other partitions via the management controller. In some embodiments, the management controller communicates with the tenants of the one or more other partitions via one of two or more sockets of a common motherboard.

[0040] According to another aspect of the disclosure, an apparatus includes one or more processors

and a non-transitory computer readable storage media storing code. The code is executable by the one or more processors to perform operations including monitoring, for faults, each of two or more partitions in a system sharing a management controller. Each of the one or more partitions is associated with a tenant. The operations include detecting a fault on a first partition of the two or more partitions and identifying an alert class of multiple alert classes for the fault. The operations include identifying that the alert class of the fault is a high-level alert class where the high-level alert class indicates that the fault affects one or more of the two or more partitions different than the first partition. The operations further include notifying the tenant of the first partition and each tenant of the one or more other partitions of the fault in response to the alert class being a high-level alert class.

[0041] In some embodiments, the high-level alert class includes multiple sub-classes. Each sub-class may be related to a response time within which a remediation action must be performed. In some embodiments, the operations further include identifying an alert sub-class corresponding to the fault in response to identifying that the alert class of the fault is a high-level alert class. In certain embodiments, the operations further include determining, based on the identified sub-class, a remediation action to be performed and/or timing of the remediation action.

[0042] In some embodiments, the operations further include, based on the identified sub-class, immediately performing, at the first partition and/or the one or more other partitions, the remediation action associated with the sub-class. In some embodiments, the operations further include scheduling, at the first partition and/or the one or more other partitions, the remediation action associated with the sub-class for a future time. In certain embodiments, the operations further include, in response to the identified alert class including the sub-class for scheduling the remediation action at a future time and in response to detecting a second fault with an identified alert class including a high-level alert class, changing the scheduled remediation to an earlier time.

[0043] In some embodiments, the operations further include building a map of hardware components related to more than one of the two or more partitions. In some embodiments, identifying the alert class the fault and identifying that the alert class is a high-level alert class further include accessing information from the map to determine if a component related to the fault is related to more than one of the two or more partitions. In some embodiments, notifying the tenants of the one or more other partitions includes communicating, over a management network, the fault to the tenants of the one or more other partitions via the management controller.

[0044] According to another aspect of the present disclosure, a system includes a partitionable motherboard with two or more partitions and a management controller configured to communicate with the two or more partitions. The management controller is connected to tenants of the two or more partitions over a management network. The system includes one or more processors on the management controller and non-transitory computer readable storage media storing code. The code is executable by the one or more processors to perform operations including monitoring, for faults, each of two or more partitions in a system sharing a management controller where each of the two or more partitions is associated with a tenant. The operations further include detecting a fault on a first partition of the two or more partitions and identifying an alert class of multiple alert classes for the fault. The operations include identifying that the alert class of the fault is a high-level alert class, where the high-level alert class indicates that the fault affects one or more other partitions of the two or more partitions different than the first partition. The operations further include notifying the tenant of the first partition and each tenant of the one or more other partitions of the fault in response to the alert class being a high-level alert class.

[0045] In some embodiments, the high-level alert class includes multiple sub-classes. Each sub-class is related to a response time within which a remediation action must be performed. The operations further include identifying an alert sub-class corresponding to the fault in response to identifying that the alert class of the fault is a high-level alert class. In some embodiments, the operations include determining, based on the identified sub-class, a remediation action to be

performed and/or timing of the remediation action.

[0046] FIG. 1 is a schematic block diagram illustrating a compute node **100** for detecting and appropriately responding to faults in a partitioned system, according to various embodiments. The compute node **100** includes a partitionable motherboard **102** having two or more partitions **112a-112n**. In some embodiments, each partition **112a-112n** corresponds to a unique tenant. In other embodiments, more than one of the partitions **112a-112n** corresponds to the same tenant. In some examples, the compute node **100** is created by partitioning the motherboard **102** with multiple CPUs **114**. In certain embodiments, each of the partitions **112a-112n** runs a different instance of an operating system (“OS”). In these and other embodiments, at least one of the partitions **112a-112n** may include one or more virtual partitions.

[0047] In some embodiments, the compute node **100** includes a management controller **104** configured to communicate with each of the partitions **112a-112n**. The management controller **104**, in some embodiments, is called a baseboard management controller (“BMC”). In some embodiments, the management controller **104** is an Xclarity® Controller (“XCC”) by Lenovo®. In other embodiments, the management controller **104** is Management Engine (“ME”) by Intel®. In other embodiments, the management controller **104** is another type by another manufacturer.

[0048] The management controller **104** is configured to communicate with a management server (not shown) over a management network. In some embodiments, the management server is an Xclarity Administrator (“XCA”) by Lenovo. In some embodiments the management server is on-site with the compute node **100**. In some embodiments, the management server is connected over a management network to an off-site management server. In some embodiments, the management network connecting the management controller **104** and the management server as well as the management network connecting the management server to the off-site management server are separate from another computer network serving the tenants of the partitions **112a-112n**. In various embodiments, the off-site management server is an XCA or an Xclarity Orchestrator (“XCO”) by Lenovo.

[0049] In some embodiments, the management controller **104** includes a fault monitor apparatus **106**, as discussed in more detail below with reference to the apparatuses **400**, **500** of FIGS. 4 and 5. In some embodiments, the management controller **104** is configured to communicate with the partitions **112a-112n** via corresponding sockets **110a-110n**. In other words, each partition **112a-112n** may be connected to a corresponding socket **110a-110n**. In some embodiments, one or more of the sockets **110a-110n** is configured to direct input/output (“I/O”) signals to or receive signals from an I/O device, such as an electronic display, a keyboard, a mouse, etc.

[0050] Each partition **112a-112n** includes at least a central processing unit (“CPU”) (labeled CPU 0 **114a** to CPU n **114n**, generically or collectively “**114**”). Each of the partitions **112a-112n** may also include any of a variety of additional components, such as memory (labeled memory **116a** to memory **116n**, generically or collectively “**116**”) and/or other resources (labeled resource **118a** to resource **118n**, generically or collectively “**118**”) such as a non-volatile storage device, a graphical processing unit (“GPU”), an accelerator, another processor, a field programmable gate array (“FPGA”), and the like. In some embodiments, two or more partitions **112a-112n** share a storage device and/or other hardware components. For clarity, the partitions **112** are depicted separate from the sockets **110**, however, the components, e.g., CPU **114**, memory **116**, resource **118**, of the partitions **112** are plugged into the sockets **110**.

[0051] FIG. 2 is a schematic block diagram illustrating a system **200** for detecting and responding to faults in a partitioned system having two tenants, according to various embodiments. The system **200** includes a compute node **100**, as described in FIG. 1, a first tenant **202**, a second tenant **204**, a computer network **206**, a FPGA **208**, remote resources **214** in the compute node **100**, a pod **216** with a switch **218**, remote hardware (“HW”) resources **220**, storage devices **222**, a pod manager **224**, and a workload orchestrator **226**, which are described below.

[0052] In FIG. 2, in some embodiments, at least one of the partitions **112a**, **112b** is associated with

a unique tenant **202**, **204**. In some embodiments, for example, a first tenant **202** is associated with a first partition **112a** and a second tenant **204** is associated with a second partition **112b**. In some embodiments, each of the first tenant **202** and the second tenant **204** is configured to communicate with its corresponding partition **112a**, **112b** via a computer network **206**. The first and second tenants **202**, **204**, in some embodiments, are clients running on a server that access the first partition **112a**, and second partition **112b** via the computer network **206**.

[0053] In some embodiments, the management controller **104** includes a fault monitor apparatus **106** configured to monitor a partition state for each of the partitions **112a**, **112b**. In some embodiments, each of the partitions **112a**, **112b** includes a CPU **114** executing an instance of an operating system. As discussed in more detail below, the fault monitor apparatus **106** may be configured to detect a fault in the first partition **112a** and/or the second partition **112b** and to determine an appropriate response to the detected fault in view of their shared components.

[0054] In some embodiments, the fault monitor apparatus **106** receives partition state information from the operating system of each partition **112a**, **112b**, where each of the partitions **112a**, **112b** includes operating system monitoring, sensors, circuits, and other equipment, data structures, etc. capable of providing partition state information. In some embodiments, partition state information is in the form of a fault and the fault monitor **106** determines an appropriate response in response to the fault. A fault may include events related to hardware components, software, processes, or the like that is indicative of a change that has occurred in the partition state of at least one of the partitions **112a**, **112b**. In some embodiments, the partition state information includes an error message, such as an error message sent to an event log.

[0055] In some embodiments, determining that there is a fault in the partition state includes determining that a monitored value exceeds a predetermined threshold value. The monitored value may include a voltage, a current, a temperature, or the like. The monitored value may be from a power supply, a CPU **114**, a compute node **100**, memory **116**, a storage device, or any other device or resource **118** in a partition **112a**, **112b** that includes an ability to monitor a parameter and compare the parameter to the predetermined threshold value.

[0056] In other embodiments, determining that a change has occurred in the system state includes the fault monitor apparatus **106** receiving a notification associated with the operating system, such as read errors, write errors, process errors, a process execution malfunction, a warning regarding memory availability, or any other notification generated by an executing operating system. In some embodiments, the notification is a predictive failure alert (“PFA”). A predictive failure alert typically monitors one or more parameters and includes one or more thresholds representative of an indication that a failure of a component is likely to occur in the near future. In other embodiments, the notification is of a failure of a hardware component of one or both of the partitions **112a**, **112b**.

[0057] In other embodiments, the notification is of an input/output error associated with input/output signals of a selected partition **112a**, **112b**. In other embodiments, the notification is of a processing error associated with a processor of the selected partition **112a**, **112b**. In other embodiments, the notification is of a crossing of a threshold where the crossing of the threshold is indicative of a problem within the selected partition **112a**, **112b**.

[0058] In some embodiments, the fault monitor apparatus **106** monitors sensors, data lines, etc. that are specifically placed for use by the fault monitor apparatus **106**. In some embodiments, the fault monitor apparatus **106** is configured to monitor various components of each partition **112a**, **112b**, such as the CPU **114**, memory **116**, various resources **118**, etc. In some embodiments, the fault monitor apparatus **106** monitors one or more workloads running on each of the partitions **112a**, **112b**.

[0059] In some embodiments, the fault monitor apparatus **106** reports faults as they occur. In other embodiments, the fault monitor apparatus **106** reports faults within a window of time. In other embodiments, the fault monitor apparatus **106** or operating system counts certain events and reports the events when a count reaches a threshold during the window of time. In other

embodiments, the fault monitor apparatus **106** stores faults in memory, which may be volatile memory or non-volatile memory. In other embodiments, the fault monitor apparatus **106** overwrites faults outside of the window of time.

[0060] In some embodiments, the fault monitor apparatus **106** interacts with the operating system (“OS”) of each partition **112a**, **112b** to collect partition state data. In other embodiments, the fault monitor apparatus **106** uses circuits, data, etc. separate from the OS of a partition **112a**, **112b** to collect partition state data. In some embodiments, the fault monitor apparatus **106** monitors the system state of the two or more partitions **112a**, **112b** using firmware running on the two or more partitions **112a**, **112b**.

[0061] In some examples, the fault monitor apparatus **106** interacts with a Unified Extensible Firmware Interface (“UEFI”) firmware or Basic Input/Output System (“BIOS”) boot firmware. UEFI firmware and BIOS boot firmware support a compute node **100** or other computing device at a low level and help boot an operating system, monitor functions, processes, etc. UEFI is a publicly available specification that defines a software interface between an operating system and platform firmware and replaces legacy BIOS boot firmware. UEFI, and to a lesser degree BIOS, support remote diagnostics and repair of computing devices with or without an operating system.

[0062] In other embodiments, the fault monitor apparatus **106** monitors the partition state of each of the two or more partitions **112a**, **112b** using a management controller **104** of a compute node **100** or similar computing device. The management controller **104**, in some embodiments, is connected to a management server over a management network. In some embodiments, the management controller **104** includes a baseboard management controller (“BMC”). In other embodiments, the management controller **104** is a datacenter secure control module (“DC-SCM”), which is discussed in detail with reference to FIG. 3 below. In some embodiments, the fault monitor apparatus **106** monitors one or more components of each of the two or more partitions **112a**, **112b**. In other embodiments, the fault monitor apparatus **106** monitors a portion of each of the two or more partitions **112a**, **112b**. One of skill in the art will recognize other ways for the fault monitor apparatus **106** to monitor for faults of each of the two or more partitions **112a**, **112b**.

[0063] In some embodiments, the partitions **112a**, **112b** to be monitored as well as what changes indicate a fault in a partition (e.g., **112a**) are configurable by a user via a graphical user interface. In other embodiments, the fault monitor apparatus **106** is configurable by a user to set thresholds and the fault monitor apparatus **106** is configurable to set logic, a hierarchy, etc. for the various thresholds. In other embodiments, the graphical user interface allows a user to determine which components, processes, workloads, etc. to monitor for partition state information. One of skill in the art will recognize other thresholds, notifications, error messages, etc. to be monitored and used by the fault monitor apparatus **106**.

[0064] In some embodiments, a field programmable gate array (“FPGA”) **208** facilitates partitioning a compute node **100** into partitions **112a**, **112b**. The FPGA **208**, in some embodiments, partitions the compute node **100** so that each partition **112a**, **112b** includes a CPU **114**, memory **116**, resources **118**, etc. The FPGA **208**, in some embodiments, configures the compute node **100** to remove one or more partitions **112a**, **112b**. In some embodiments, the FPGA **208** facilitates communication between the fault monitor apparatus **106** and the partitions **112a**, **112b**. In various embodiments, the FPGA **208** differs from a remote hardware resource **220** that is an FPGA used for execution of workloads.

[0065] In some embodiments, a workload orchestrator **226** coordinates execution of workloads on partitions **112a**, **112b** of one or more compute nodes **100**. In some embodiments, the workload orchestrator **226** coordinates with the FPGA **208** to create partitions **112a**, **112b** that include virtual machines (“VMs”), containers, etc. for various tenants **202**, **204** in communication with the one or more compute nodes **100** and partitions **112a**, **112b** over a computer network **206**.

[0066] In some embodiments, the partitions **112a**, **112b** also include at least one remote resource **214** which emulates a remote hardware resource **220** located external to the compute node **100**. The

remote hardware resources **220**, in some embodiments are in a pod **216** that includes various remote resources **214**, such as accelerators, FPGAs, storage devices, CPUs, GPUs, and the like. The pod **216** typically includes a switch **218** configured to connect the remote hardware resources **220** to compute nodes **100** as directed by a pod manager **224**. The fault monitor apparatus **106** is configured, in some embodiments, to monitor faults of a remote hardware resource **220** that is emulated by a remote resource **214** of a partition **112a**, **112b**.

[0067] In some embodiments, the compute nodes **100** include multiple sockets that each accommodate a CPU **114** where each CPU **114** is supported by memory **116** and other components. A common configuration of a compute node **100** includes two sockets **110a**, **110n**, as depicted in FIG. 1.

[0068] The compute nodes **100** communicate with an associated tenant **202**, **204**, respectively, over a computer network **206** that may include a LAN, a WAN, a fiber network, a wireless connection, the Internet, etc. and may include multiple networks. The wireless connection may be a mobile telephone network. The wireless connection may also employ a Wi-Fi network based on any one of the Institute of Electrical and Electronics Engineers (“IEEE”) 802.11 standards. Alternatively, the wireless connection may be a BLUETOOTH® connection. In addition, the wireless connection may employ a Radio Frequency Identification (“RFID”) communication including RFID standards established by the International Organization for Standardization (“ISO”), the International Electrotechnical Commission (“IEC”), the American Society for Testing and Materials® (“ASTM”®), the DASH7™ Alliance, and EPCGlobal™.

[0069] Alternatively, the wireless connection may employ a ZigBee® connection based on the IEEE 802 standard. In one embodiment, the wireless connection employs a Z-Wave® connection as designed by Sigma Designs®. Alternatively, the wireless connection may employ an ANT® and/or ANT+® connection as defined by Dynastream® Innovations Inc. of Cochrane, Canada.

[0070] The wireless connection may be an infrared connection including connections conforming at least to the Infrared Physical Layer Specification (“IrPHY”) as defined by the Infrared Data Association® (“IrDA”®). Alternatively, the wireless connection may be a cellular telephone network communication. All standards and/or connection types include the latest version and revision of the standard and/or connection type as of the filing date of this application.

[0071] FIG. 3 is a schematic block diagram illustrating another embodiment of a system **300** for detecting and appropriately responding to faults in a partitioned system, where the partitions **112a**, **112b** are in a partitioned compute node **100**, according to various embodiments. In some embodiments, the partitions **112a**, **112b** are in a composable system. The system **300** includes a compute node **100** having a management controller **104** and a fault monitor apparatus **106** substantially similar to those described above in relation to the systems **100**, **200** of FIGS. 1 and 2. In the embodiments of the system **300** of FIG. 3, the management controller **104** is in the form of a datacenter secure compute module (“DC-SCM”) **304** and includes the fault monitor apparatus **106**.

[0072] In the depicted embodiments, the fault monitor apparatus **106** is connected to each of two partitions **112a**, **112b**. The system **300** includes two partitions **112a**, **112b**, labeled zone 1 and zone 2, which are substantially similar to those described above in relation to the system **200** of FIG. 2. Each partition **112a**, **112b** includes a CPU **114**, memory **116**, and a resource **118** as in the system **200** of FIG. 2. The partitions **112a**, **112b** may also include other installed resources **118**. The compute node **100** also includes a zone 3 **310** which includes an I/O switch **308**. In some embodiments, zone 3 **310** is partitionable into one or two parts, depending on whether or not I/O devices **306a**, **306b** are separated to service each of the first partition **112a** and the second partition **112b** or if the I/O switch **308** is used to connect to a single partition **112a**, **112b** at a time. In some embodiments, the I/O switch **308** is a keyboard, video, mouse (“KVM”) switch. The I/O switch **308**, in some embodiments, includes various controllers, such as a GPU, a universal serial bus (“USB”) controller, etc. In other embodiments, the I/O switch **308** switches I/O signals and another device includes I/O controllers, a GPU, etc.

[0073] The fault monitor apparatus **106** may be configured to monitor the I/O switch **308**. In some embodiments, the I/O switch **308** is configured to switch input of the I/O switch **308** to send data for display of one or more elements of the selected partition (e.g., **112a**) to an electronic display, for example.

[0074] The system **300** of FIG. 3 includes a DC-SCM **304**. The DC-SCM **304** is a management controller **104** compliant with an open source specification. The DC-SCM **304** complies with a DC-SCM specification, such as the DC-SCM 2.0 specification. The DC-SCM **304**, in some embodiments, includes a baseboard management controller (“BMC”) and is connected to a management network. Typically, a DC-SCM **304** includes a BMC plus other supporting components, such as memory, buses, bus controllers, a GPU, and the like. In some embodiments, the DC-SCM **304** is a card in a dedicated slot on a motherboard of the compute node **100** where the slot complies with a DC-SCM specification.

[0075] As described above, the management network, in some embodiments, is a network different than the computer network **206** used for communication with clients, tenants **202**, **204**, etc. for communication of data from workloads running on the compute node **100**, etc. In other embodiments, the management network uses a same computer network **206** as is used for data from workloads but runs securely.

[0076] In some embodiments, the fault monitor apparatus **106** is distributed between a motherboard **102** of the compute node **100** and the DC-SCM **304**. In some examples, one or more I/O switches **308** are on the motherboard **102** and the fault monitor apparatus **106** is in the DC-SCM **304**. In other embodiments, components of the fault monitor apparatus **106** are distributed differently.

[0077] In the embodiments of FIG. 3, the compute node **100** is connected to other devices, such as storage **302a**, **302b**, I/O devices **306a**, **306b**, which may be installed through cables, in slots, etc. The storage **302a**, **302b**, I/O devices **306a**, **306b**, etc., in some embodiments, are situated to service each CPU **114** when partitioned. In other embodiments, I/O devices **306a**, **306b**, such as an electronic display, keyboard, mouse, etc. are connected through the DC-SCM **304**, either locally or remotely, to provide system administrator access to the compute node **100** and partitions **112a**, **112b** of the compute node **100**.

[0078] FIG. 4 is a schematic block diagram illustrating an apparatus **400** for monitoring and responding to faults in a partitioned system **100**, **200**, **300**, according to various embodiments. The apparatus **400** includes a fault monitor **106** with a fault monitor module **402**, a class identifier (“ID”) module **404**, and a notification module **406**, which are described below. In some embodiments, the fault monitor **106** monitors two or more partitions **112a**, **112b** for faults. The fault monitor apparatus **106** may be located in a management controller **104**, in a DC-SCM **304**, on a motherboard **102** of a compute node **100**, or the like. In some embodiments, the apparatus **400** is implemented using code stored on a computer readable storage media. The code is executable on a processor. In other embodiments, all or a portion of the apparatus **400** is implemented using hardware circuits and/or a programmable hardware device.

[0079] In some embodiments, the apparatus **400** includes a fault monitor module **402** configured to monitor, for faults, each of two or more partitions **112a**, **112b** in a partitioned system **200**. The partitions **112a**, **112b** share a management controller **104** and each of the partitions **112a**, **112b** is associated with a tenant **202**, **204**. In various embodiments, the fault monitor module **402** monitors the partitions **112** for faults as described above in relation to the system **200** of FIG. 2. In some embodiments, the fault monitor module **402** detects faults based on information sent by the partitions **112** to the management controller **104** or DC-SCM **304**. In other embodiments, the fault monitor module **402** is configured to receive fault information from an FPGA **208** relaying information from the partitions **112**. The various types of faults that may be monitored by the fault monitor module **402** are described above.

[0080] The fault monitor module **402** is also configured to detect a fault on a first partition **112a** of the two or more partitions **112a**, **112b**. The fault may include, for example, a power fault, a thermal

event, a hardware component failure, a catastrophic software error, and/or the like. The detected fault may physically impact a partition **112a**, **112b** different than the partition **112a**, **112b** where the detected fault occurred.

[0081] The apparatus **400** includes a class identification module **404** configured to identify an alert class of a plurality of alert classes **600**. The alert class corresponds to the detected fault. In some embodiments, multiple alert classes **600** correspond to one or more faults. In some embodiments, as shown in FIG. **6**, the alert classes **600** are hierarchically arranged to include a high-level alert class **602**, a medium-level alert class **610**, and a low-level alert class **612**. The high-level alert class **602** indicates that the fault affects at least one partition **112a**, **112b** other than the partition **112a**, **112b** in which the fault occurred. In some embodiments, the medium-level alert class **610** indicates that the fault affects only the partition **112a**, **112b** in which the fault occurred. In certain embodiments, the low-level alert class **612** indicates that the fault is not a significant problem and affects only the partition **112a**, **112b** in which the fault occurred.

[0082] In certain embodiments, one or more of the alert classes **600** is associated with one or more sub-classes **604**. Each sub-class **604** may identify a remediation action corresponding to the fault and/or a response time within which the remediation action must be performed. In some embodiments, the high-level alert class **602** includes multiple sub-classes **604** where each sub-class (e.g., **606**, **608**) of the plurality of sub-classes **604** is related to a response time within which a remediation action must be performed and the class identification module **404** is configured to identify an alert sub-class **604** corresponding to the detected fault in response to identifying that the alert class **600** of the fault is a high-level alert class **602**. In some embodiments, the sub-classes **604** include an urgent sub-class **606** to identify faults associated with a remediation action that must be performed immediately. In some embodiments, the sub-classes **604** include a non-urgent sub-class **608** to identify faults associated with a non-urgent remediation action that may be scheduled for a future time. In some embodiments, the sub-classes **604** of the high-level alert classes **602** include other sub-classes **604** not depicted in FIG. **6** associated with other remediation timing. In certain embodiments the non-urgent sub-class **608** identifies a fault that may be simply logged or recorded.

[0083] The alert classes **600** of FIG. **6** are only illustrative of some embodiments of alert classes **600**. The fault detection apparatus **106** described herein contemplates other fault classes, types, etc. and is configured to identify faults that affect more than one partition **112** of a partitionable system **100**, **200**, **300**. One of skill in the art will recognize various alert classes that affect more than one partition **112**.

[0084] The class identification module **404** is also configured to identify that the alert class of the fault is a high-level alert class **602** where the high level alert class indicates that the fault affects one or more other partitions (e.g., **112b**) of the two or more partitions **112a-112n** different than the first partition **112a**. In other embodiments, the class identification module **404** identifies that the detected fault is a medium-level alert class **610** or a low-level alert class **612**.

[0085] The apparatus **400** includes a notification module **406** configured to notify the tenant **202** of the first partition **112a** and each tenant of the one or more other partitions (e.g., **112b-112n**) of the fault in response to the alert class being the high-level alert class. While the examples described herein include the fault occurring in the first partition **112a**, other embodiments are contemplated where the fault occurs in any of the other partitions **112b-112n** and the notification module **406** notifies partitions **112** where the fault did not occur when the alert class corresponds to a high-level alert class **602** affecting more than one partition **112**. In some embodiments, the notification module **406** is configured to notify one or more tenants **204** of other partitions **112a**, **112b** if fault monitor apparatus **106** determines that the fault is likely to affect any partitions **112a**, **112b** other than the first partition **112a**. In some embodiments, the notification module **406** notifies the tenant **202** of the first partition **112a** and the tenants **204** of the additional partitions **112b** if the identified alert class **600** is a high-level alert class **602**. In some embodiments, the notification module **406** notifies the tenants **204** of the other partitions **112b** via the management controller **104**. In some

embodiments, the management controller **104** communicates with the tenants **202**, **204** via a management network. In other embodiments, the notification module **406** notifies partitions **112** using a communication channel other than the management network.

[0086] FIG. 5 is a schematic block diagram illustrating another apparatus **500** for monitoring and responding to faults in a partitioned system **100**, **200**, **300**, according to various embodiments. The apparatus **500** includes another fault monitor apparatus **106** that includes a fault monitor module **402**, a class identification module **404**, and a notification module **406**, which are substantially similar to those describe above in relation to the apparatus **400** of FIG. 4. In various embodiments, the fault monitor apparatus **106** includes a remediation module **502**, a remediation timing module **504**, and/or a mapping module **506**, which are described below. In some embodiments, the apparatus **500** is implemented similar to the apparatus **400** of FIG. 4.

[0087] In some embodiments, the apparatus **500** includes a remediation module **502** configured to identify a remediation action to be performed and/or timing of the remediation action based on the identified alert class **600** of the detected fault. In some embodiments, the remediation module **502** further identifies one or more remediation actions associated with each sub-class **604**. Remediation actions may include, for example, logging the fault, moving data, shedding load, refusing critical loads, and/or redirecting new data to a non-shared drive.

[0088] In some embodiments, the remediation actions are designed to prepare a partition (e.g., **112b**) based on faults with alert classes that affect partitions (e.g., **112b-112n**) different than the partition **112a** where the fault occurred. For example, where the fault is of a type where the entire motherboard **102** of a compute node **100** is shut down, remediation actions for the non-fault partitions **112b-112n** may require shedding load, refusing workloads, moving data, or other actions commensurate with shutting down the non-fault partitions **112b-112n**. In other embodiments, the fault may be associated with a first storage device **222a** that is used by the first partition **112a** where the fault was detected as well a second partition **112b** so that a remediation action affects the second partition **112b**. The remediation action may then be to move data used by the second partition **112b** to a second storage device **222b** in preparation for a remediation action affecting the first storage device **222a**.

[0089] In other examples, the detected fault of the first partition **112a** may include rebooting the compute node **100**, which affects the other partitions **112b-112n** so that the remediation module **502** is configured to take remediation actions on the other partitions **112b-112n** in preparation for the reboot. In other examples, the fault detected on the first partition **112a** may involve a power supply providing power to the first partition **112a** as well as to one or more of the other partitions **112b-112n** so that the remediation module **502** includes remediation actions for the other partitions **112b-112n** in addition to remediation actions for the first partition **112a**. One of skill in the art will recognize other remediation actions for the other partitions **112b-112n** in addition to remediation actions for the first partition **112a** where the fault occurred.

[0090] In various embodiments, the remediation actions for the various alert classes **600** and sub-classes **604** are appropriate for the detected fault. In some embodiments, the remediation action is more than the notifications transmitted by the notification module **406**. In some embodiments, the remediation module **502** initiates the remediation action for a particular fault automatically without intervention of a system administrator or other user. In other embodiments, the remediation module **502** initiates remediation actions after approval by a system administrator or user where the remediation module **502** may work in conjunction with the notification module **406** to identify a remediation action in the notification sent to a system administrator or user.

[0091] In some embodiments, the apparatus **500** includes a remediation timing module **504** configured to associate a response time, response window, duration of time, etc. for response with each alert class **600** and/or sub-class **604**. In some embodiments, the high-level alert class **602** includes an urgent sub-class **606** and a non-urgent sub-class **608**. In some embodiments, the remediation timing module **504** requires immediate performance of a remediation action if the alert

class **600** is a high-level alert class **602** with an urgent sub-class **606**. In other embodiments, the remediation timing module **504** defers a remediation action where the alert class **600** is a high-level alert class **602** and the sub-class **604** is a non-urgent sub-class **608** or other sub-class **604** that is not urgent. For example, a non-urgent sub-class **608** may include a remediation action that occurs when a technician replaces a part, performs maintenance, or the like. In some embodiments, the remediation timing module **504** schedules some remediation action immediately, such as refusing critical workloads, while deferring other remediation actions, such as shedding all load until a later time, such as when a technician is about to replace a part, shut down the motherboard **102**, or the like.

[0092] In certain embodiments, the remediation timing module **504** schedules performance of the remediation action immediately or for a future time for the first partition **112a** where the fault occurred if the alert class **600** is a medium-level alert class **610** or a low-level alert class **612**. In some embodiments, the medium-level alert class **610** and/or the low-level alert class **612** include sub-classes indicative of timing of remediation actions associated with the first partition **112a** where the fault occurred. In some embodiments, the remediation timing module **504** is configured to change or move a scheduled future remediation time to an earlier time if a second fault is detected in one of the partitions **112a**, **112b** and the second fault belongs to the high-level alert class **602** and/or is of a nature where a remediation action should occur before a scheduled remediation action.

[0093] In some embodiments, the apparatus **500** includes a mapping module **506** configured to build a map of hardware components corresponding to the partitions **112a**, **112b** of a partitioned compute node **100**. In some embodiments, the fault monitor apparatus **106** builds the map based on configuration mappings of each partition **112a**, **112b**. In some embodiments, the map includes hardware components shared between two or more of the partitions **112a**, **112b**. In some examples, two or more of the partitions e.g., **112a**, **112b** include one or more shared storage devices **222**. In other embodiments, two or more buses or PCIe adapters are assigned to separate partitions **112a**, **112b**, but may share a PCIe riser card. In some embodiments, the fault monitor apparatus **106** defines the hierarchy of alert classes **600** based upon the map of hardware components.

[0094] FIG. 7 is a schematic flow chart diagram illustrating a method **700** for monitoring and responding to faults in a partitioned system **100**, **200**, **300** sharing a management controller **104** according to various embodiments. The method **700** begins and monitors **702**, for faults, each of two or more partitions **112a-112n** in a system **100**, **200**, **300** sharing a management controller **104** (or DC-SCM **304**). In some embodiments, each of the two or more partitions **112a-112n** includes a CPU **114** executing an instance of an operating system. Each partition **112a-112n** is associated with a tenant **202**, **204**, etc.

[0095] The method **700** detects **704** a fault on a first partition **112a** of the two or more partitions **112a-112n** and identifies **706** an alert class **600** of a plurality of alert classes **600** for the fault. The method **700** determines **708** whether the alert class **600** is a high-level alert class **602**. The high-level alert class indicates that the fault affects one or more other partitions e.g., **112b-112n** of the two or more partitions **112a-112n** different than the first partition **112a**. If the method **700** determines **708** that that alert class **600** is a high-level alert class **602**, the method **700** notifies **710** the tenant **202** of the first partition **112a** and each tenant **204**, etc. of the one or more other partitions **112b-112n** of the fault, and the method **700** ends. If the method **700** determines **708** that the alert class **600** is not a high-level alert class **602**, the method **700** alerts **712** the tenant **202** of the first partition **112a** only, and the method **700** ends. In various embodiments, all or a portion of the method **700** is implemented using the fault monitor module **402**, the class identification module **404**, and/or the notification module **406**.

[0096] FIG. 8 is a schematic flow chart diagram illustrating another method **800** for monitoring and responding to faults in a partitioned system **100**, **200**, **300** sharing a management controller **104** according to various embodiments. The method **800** begins and builds **802** a map of hardware

components in a partitioned system **100, 200, 300**. The method **800** monitors **804** two or more partitions **112a-112n** for faults and detects **806** a fault in one of the partitions **112a-112n** and identifies **808** an alert class **600** for the fault. The method **800** determines **810** if the alert class **600** is a high-level alert class **602**. If the method **800** determines **810** that the alert class **600** is not a high-level alert class **602**, the method **800** alerts **812** the tenant (e.g., **202**) of the partition (e.g., **112a**), corresponding to the fault, and the method **800** ends.

[0097] If the method **800** determines **810** that the alert class **600** is a high-level alert class **602**, the method **800** identifies **814** a sub-class **604** corresponding to the fault and determines **816** at least one remediation action for the fault based on the sub-class **604**. The method **800** determines **818** whether the sub-class **604** is an urgent sub-class **606**. If the method **800** determines **818** that the sub-class **604** is an urgent sub-class **606**, the method **800** immediately performs **820** the remediation action, and the method **800** ends. If the method **800** determines **818** that the sub-class **604** is a non-urgent sub-class **608**, the method **800** schedules **822** the remediation action at a future time or within a future time frame.

[0098] The method **800** determines **824** whether a second fault is detected within the partitioned system **100, 200, 300**, which may also include determining if the second fault also belongs to a high-level alert class **602**. If the method **800** determines **824** that there is a second fault that is a high-level alert class **602**, for some faults the method **800** changes **826** the scheduled time for performing the remediation action to an earlier time, or re-schedules the remediation action for immediate performance, and the method **800** ends. If the method **800** determines **824** that there is no fault that is a high-level fault before the scheduled remediation action, the method **800** ends after the remediation action. In various embodiments, all or a portion of the method **800** is implemented using the fault monitor module **402**, the class identification module **404**, the notification module **406**, the remediation module **502**, the remediation timing module **504**, and/or the mapping module **506**.

[0099] Embodiments may be practiced in other specific forms. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A method comprising: monitoring, for faults, each of two or more partitions in a system sharing a management controller, wherein each of the two or more partitions is associated with a tenant; detecting a fault on a first partition of the two or more partitions; identifying an alert class of a plurality of alert classes for the fault; identifying that the alert class of the fault is a high-level alert class, the high-level alert class indicating that the fault affects one or more other partitions of the two or more partitions different than the first partition; and notifying the tenant of the first partition and each tenant of the one or more other partitions of the fault in response to the alert class being the high-level alert class.
2. The method of claim 1, wherein the high-level alert class comprises a plurality of sub-classes, wherein each sub-class of the plurality of sub-classes is related to a response time within which a remediation action must be performed, and further comprising identifying an alert sub-class corresponding to the fault in response to identifying that the alert class of the fault is a high-level alert class.
3. The method of claim 2, further comprising determining, based on the identified sub-class, a remediation action to be performed and/or timing of the remediation action.
4. The method of claim 3, wherein the remediation action comprises, for the first partition and the one or more other partitions, moving data, shedding load, refusing critical loads, and redirecting

new data to a non-shared drive.

5. The method of claim 3, further comprising, based on the identified sub-class, immediately performing, at the first partition and/or the one or more other partitions, the remediation action associated with the sub-class.
6. The method of claim 3, further comprising, based on the identified sub-class, scheduling, at the first partition and/or the one or more other partitions, the remediation action associated with the sub-class for a future time.
7. The method of claim 6, wherein in response to the identified alert class comprising the sub-class for scheduling the remediation action at a future time and in response to detecting a second fault with an identified alert class comprising a high-level alert class, further comprising changing the scheduled remediation to an earlier time.
8. The method of claim 1, further comprising building a map of hardware components related to more than one of the two or more partitions, wherein identifying the alert class of the plurality of alert classes for the fault and identifying that the alert class is a high-level alert class further comprises accessing information from the map to determine if a component related to the fault is related to more than one of the two or more partitions.
9. The method of claim 1, wherein notifying the tenants of the one or more other partitions comprises communicating, over a management network, the fault to the tenants of the one or more other partitions via the management controller.
10. The method of claim 9, wherein the management controller communicates with the tenants of the one or more other partitions via one of two or more sockets of a common motherboard.
11. An apparatus comprising: one or more processors; and non-transitory computer readable storage media storing code, the code being executable by the one or more processors to perform operations comprising: monitoring, for faults, each of two or more partitions in a system sharing a management controller, wherein each of the one or more partitions is associated with a tenant; detecting a fault on a first partition of the two or more partitions; identifying an alert class of a plurality of alert classes for the fault; identifying that the alert class of the fault is a high-level alert class, the high-level alert class indicating that the fault affects one or more other partitions of the two or more partitions different than the first partition; and notifying the tenant of the first partition and each tenant of the one or more other partitions of the fault in response to the alert class being a high-level alert class.
12. The apparatus of claim 11, wherein the high-level alert class comprises a plurality of sub-classes, wherein each sub-class of the plurality of sub-classes related to a response time within which a remediation action must be performed, and further comprising identifying an alert sub-class corresponding to the fault in response to identifying that the alert class of the fault is a high-level alert class.
13. The apparatus of claim 12, further comprising determining, based on the identified sub-class, a remediation action to be performed and/or timing of the remediation action.
14. The apparatus of claim 13, further comprising, based on the identified sub-class, one of: immediately performing, at the first partition and/or the one or more other partitions, the remediation action associated with the sub-class; and scheduling, at the first partition and/or the one or more other partitions, the remediation action associated with the sub-class for a future time.
15. The apparatus of claim 14, wherein in response to the identified alert class comprising the sub-class for scheduling the remediation action at a future time and in response to detecting a second fault with an identified alert class comprising a high-level alert class, further comprising changing the scheduled remediation to an earlier time.
16. The apparatus of claim 11, further comprising building a map of hardware components related to more than one of the two or more partitions, wherein identifying the alert class of the plurality of alert classes for the fault and identifying that the alert class is a high-level alert class further comprises accessing information from the map to determine if a component related to the fault is

related to more than one of the two or more partitions.

17. The apparatus of claim 11, wherein notifying the tenants of the one or more other partitions comprises communicating, over a management network, the fault to the tenants of the one or more other partitions via the management controller.

18. A system comprising: a partitionable motherboard with two or more partitions; a management controller configured to communicate with the two or more partitions, the management controller connected to tenants of the two or more partitions over a management network; one or more processors on the management controller; and non-transitory computer readable storage media storing code, the code being executable by the one or more processors to perform operations comprising: monitoring, for faults, each of two or more partitions in a system sharing a management controller, wherein each of the two or more partitions is associated with a tenant; detecting a fault on a first partition of the two or more partitions; identifying an alert class of a plurality of alert classes for the fault; identifying that the alert class of the fault is a high-level alert class, the high-level alert class indicating that the fault affects one or more other partitions of the two or more partitions different than the first partition; and notifying the tenant of the first partition and each tenant of the one or more other partitions of the fault in response to the alert class being a high-level alert class.

19. The system of claim 18, wherein the high level alert class comprises a plurality of sub-classes, wherein each sub-class of the plurality of sub-classes related to a response time within which a remediation action must be performed, and further comprising identifying an alert sub-class corresponding to the fault in response to identifying that the alert class of the fault is a high level alert class.

20. The system of claim 19, further comprising determining, based on the identified sub-class, a remediation action to be performed and/or timing of the remediation action.
